Let’s attempt to understand it.

Time is not what we think it is

That the nature of time is different from the common idea which we have of it was already clear over a century ago. Special and general relativity made this explicit. Today, the inadequacy of our common-sense view of time can be easily verified in a laboratory.

Let’s reconsider, for example, the first consequence of general relativity, as illustrated in Chapter 3. Take two watches, ensure that they mark exactly the same time, place one on the floor and the other on a piece of furniture. Wait for about half an hour and then bring them back next to each other. Will they still tell the same time?

As described in Chapter 3, the answer is no. The watches which we usually wear on our wrists, or have on our mobile phones, are not sufficiently precise to allow us to verify this fact, but in physics laboratories all over the world there are timepieces precise enough to demonstrate the discrepancy which occurs: the watch left on the floor is slow when compared to the one which has been raised above it.

Why? Because time does not pass in the same way everywhere in the world. In some places, it flows more quickly; in others, more slowly. The closer you get to the Earth, where gravity[fn39] is more intense, the slower time passes. Remember the twins in Chapter 3, who ended up with different ages as a result of having lived one at sea level and one in the mountains? The effect is very slight: the time gained during a life spent by the sea, with respect to one passed in the mountains, consists of fractions of a second – but the smallness of the amount does not alter the fact that there is a real difference. Time does not work as we customarily imagine it does.

We must not think of time as if there were a great cosmic clock that marks the life of the universe. We have known for more than a century that we must think of time instead as a localized phenomenon: every object in the universe has its own time running, at a pace determined by the local gravitational field.

But even this notion of a localized time no longer works when we take the quantum nature of the gravitational field into account. Quantum events are no longer ordered by the passage of time at the Planck scale. Time, in a sense, ceases to exist.

What does it mean to say that time does not exist?

First, the absence of the variable time from the fundamental equations does not imply that everything is immobile and that change does not happen. On the contrary, it means that change is ubiquitous. Only: elementary processes cannot be ordered along a common succession of instants. At the extremely small scale of the quanta of space, the dance of nature does not develop to the